

Kazakh-British Technical University

Databases

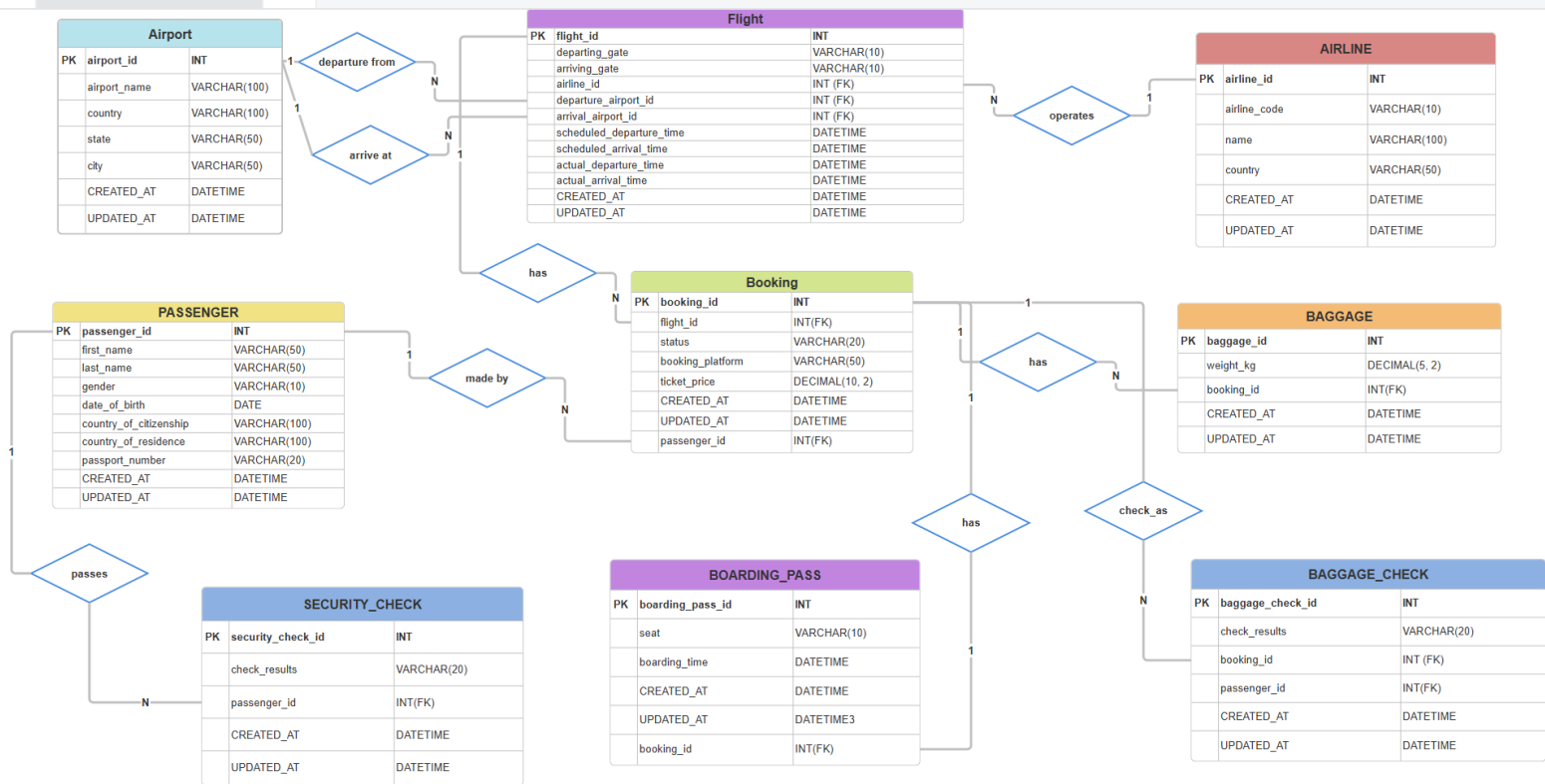
Laboratory work № 1

ERD diagram.

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Legend / Notation



Relationship



Entity (table)

1 N

Relationship line



Primary key



Foreign key

1 - one

N - many

1 - 1 - exactly one

0..1 - zero or one

1..* - one or many

0..* - zero or many

Entities and their Attributes

Airport

Stores information about airports (id, name, country, state, city).

Flight

Stores flight schedule and actual times, gates, and references to airline and airports.

Airline

Stores information about airlines.

Booking

Stores booking details for flights.

Boarding Pass

Stores boarding pass information for each booking.

Baggage

Stores baggage information for each booking.

Baggage Check

Stores results of baggage inspection at the airport.

Passenger

Stores personal information about passengers.

Security Check

Stores results of security checks for passengers.

Relationship

1. AIRPORT - FLIGHT (departs from)

One airport can be the departure airport for many flights.
Each flight has one departure airport.
Cardinality: 1:N Participation: Airport - partial Flight - total

2. AIRPORT - FLIGHT (arrives at)

One airport can be the arrival airport for many flights.
Each flight has one arrival airport.
Cardinality: 1:N Participation: Airport - partial Flight - total

3. AIRLINE - FLIGHT (operates)

One airline can operate many flights.
Each flight is operated by one airline.
Cardinality: 1:N Participation: Airline - partial Flight - total

4. FLIGHT - BOOKING (has)

One flight can have many bookings.
Each booking is for one flight.
Cardinality: 1:N Participation: Flight - partial Booking - total

5. PASSENGER - BOOKING (made by)

One passenger can make many bookings.
Each booking is made by one passenger.
Cardinality: 1:N Participation: Passenger - partial Booking - total

6. BOOKING - BOARDING_PASS (has)

One booking can have 1 boarding passes.
Each boarding pass belongs to one booking
Cardinality: 1:1 Participation: Booking - partial Boarding Pass - total

7. BOOKING - BAGGAGE (has)

One booking can have many baggage items.
Each baggage item belongs to one booking.
Cardinality: 1:N Participation: Booking - partial Baggage - total

8. PASSENGER - SECURITY_CHECK (passes)

One passenger can have many security checks.
Each security check belongs to one passenger.
Cardinality: 1:N Participation: Passenger - partial Security Check - total

9. BAGGAGE - BAGGAGE_CHECK (checked as)

One baggage item can have many checks over time.
Each baggage check is for one baggage item.
Cardinality: 1:N Participation: Baggage - partial Baggage Check - total